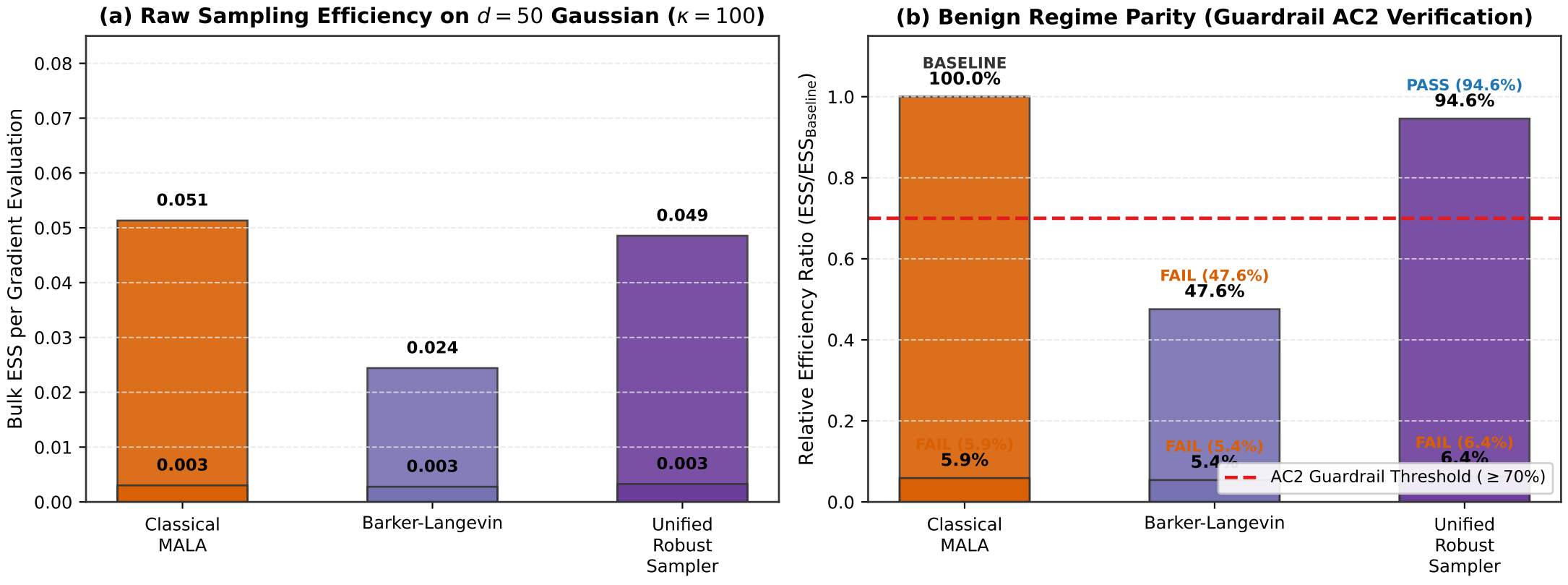
(a) Raw Sampling Efficiency on $d = 50$ Gaussian ($\kappa = 100$)
Bulk ESS per Gradient Evaluation
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0.051
0.003
0.024
0.003
0.049
0.003
Classical MALA
Barker-Langevin
Unified Robust Sampler
(b) Benign Regime Parity (Guardrail AC2 Verification)
Relative Efficiency Ratio (ESS/ESS$_{\text{Baseline}}$)
1.0
0.8
0.6
0.4
0.2
0.0
BASELINE
100.0%
FAIL (5.9%)
5.9%
FAIL (47.6%)
47.6%
FAIL (5.4%)
5.4%
PASS (94.6%)
94.6%
FAIL (6.4%)
6.4%
AC2 Guardrail Threshold (≥ 70%)
Classical MALA
Barker-Langevin
Unified Robust Sampler